

Hackathon Problem Statements

Regd No: 25091A32J5, 25091A32A1

Problem Statement No: 31

Epidemic Contact-Tracing Simulation Build a contact-tracing simulation using union-find (disjoint set) data structures to efficiently group individuals into connected clusters of potential exposure during an epidemic. Combine with basic probability models to estimate infection spread likelihood within each cluster. Use OOP to represent individuals, locations, and exposure events. This demonstrates how digital contact-tracing systems, like those used during COVID-19, can be built and scaled efficiently.

Regd No: 25091A32J0, 25091A32H2

Problem Statement No: 39

Parking Slot Allocation System Build a smart parking allocation system using greedy and priority-based scheduling algorithms to assign available slots to incoming vehicles based on arrival time, duration, and vehicle type. Apply demand forecasting using historical occupancy data to predict peak parking congestion periods. Use OOP to model parking lots, slots, and vehicles with booking states. This reduces the time and fuel wasted circling for parking in congested urban commercial areas.

Regd No: 25091A32H9, 25091A32P1

Problem Statement No: 43

City-Wide Crime Hotspot Predictor Create a crime hotspot prediction tool using spatial clustering (DBSCAN) on historical, anonymized incident location and time datasets to identify high-risk zones and time windows. Visualize results as heatmaps to support smarter police patrol allocation. Use OOP to structure incident records, zones, and patrol units. This data-driven approach can improve resource deployment efficiency for urban police departments with limited patrol capacity.

Regd No: 25091A32A5, 25091A3290

Problem Statement No: 77

Urban Tree Canopy Mapping & Green Cover Analyzer Build a green cover analysis tool using clustering algorithms on GIS/satellite-derived vegetation index data to map tree canopy density across city zones and identify areas needing afforestation. Use OOP to model zones, canopy metrics, and planting recommendations. This supports urban forestry departments in data-driven green cover expansion planning to combat urban heat islands.

Regd No: 25091A3231, 25091A3240

Problem Statement No: 96

Stock Market Trend Predictor Design a stock trend analysis tool using time-series regression and moving-average techniques on historical stock price datasets to identify trend patterns and generate simple buy/hold/sell signals for educational purposes. Use OOP to model stocks, price histories, and trading strategies. This helps students understand basic quantitative finance concepts through hands-on data analysis rather than pure theory.

Regd No: 25091A32B4, 25091A32C5

Problem Statement No: 104

Financial Literacy Quiz Adaptive Engine Design an adaptive financial literacy quiz system using decision-tree-based difficulty adjustment that personalizes question difficulty based on a learner's ongoing performance, similar to adaptive testing platforms. Use OOP to model quiz modules, questions, and learner progress. This supports scalable financial literacy education for populations with varying baseline knowledge levels.